A22 | Linear Algebra

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1 Complex Numbers

1.1 Cartesian

Definition.

$z = a + bi \quad w = c + di$ for $a, b, c, d \in \mathbb{R}$

- $a = Re(z) = r\cos\theta$ is the **real** part of z
- $b = Im(z) = r\sin\theta$ is the **imaginary** part of z
- $z = w$ iff $a = c \wedge b = d$

1.2 Polar

Definition.

$z = re^{i\theta} \quad w = se^{i\varphi}$ for $r, s, \theta, \varphi \in \mathbb{R}$

- $r = |z| = \sqrt{a^2 + b^2}$ is the **modulus** of z
- $\theta = \arg z = \operatorname{atan}(b/a)$ is **an argument** of z
  $\theta \in (-\pi, \pi]$ is **the principle argument** of z
  $\operatorname{atan}(-b/a)$ and $\operatorname{atan}(b/-a)$ are not the same for $a, b > 0$
- $z = re^{i(\theta + 2\pi k)}$ is the **general solution**

1.3 Add

Definition. $z + w = (a + c) + (c + d)i$

- commutativity, associativity
- add-identity: 0
- add-inverse: $-a - bi = -re^{i\theta}$

1.4 Mul

Definition. $zw = rse^{i(\theta + \varphi)}$

- commutativity, associativity, distributivity
- mul-identity: 1
- mul-inverse: $\dfrac{1}{z} = \dfrac{\overline{z}}{|z|^2}$

1.5 Conjugate

Definition. $\overline{z} = a - bi$

- $\overline{z \pm w} = \overline{z} \pm \overline{w}$
- $\overline{zw} = \overline{z}\overline{w}$
- $\overline{(\overline{z})} = z$
- $\overline{z} = z$ iff $z \in \mathbb{R}$
- $z\overline{z} = |z|^2 = r^2 = a^2 + b^2$

1.6 Modulus

Definition.

$|z| = \sqrt{a^2 + b^2} = r$

- $|z|^2 = z\overline{z} = a^2 + b^2$
- $|zw| = |z||w|$
- $|z + w| \leq |z| + |w|$ (Δ inequality)
- $|z - w| = \sqrt{(a-c)^2 + (b-d)^2}$ (distance)

Example. Prove $|z| \leq |a| + |b| \leq \sqrt{2}|z|$ for $z = a + bi \in \mathbb{C}$

$$\begin{aligned}
|z| &\leq |a| + |b| \\
|z|^2 &\leq |a|^2 + |b|^2 + 2|a||b| \\
a^2 + b^2 &\leq a^2 + b^2 + 2|ab| \\
0 &\leq |ab| \\
|a| + |b| &\leq \sqrt{2}|z| = \sqrt{2a^2 + 2b^2} \\
|a|^2 + |b|^2 + 2|a||b| &\leq 2a^2 + 2b^2 \\
0 &\leq |a|^2 + |b|^2 - 2|a||b| \\
0 &\leq (|a| - |b|)^2
\end{aligned}$$

Example. Prove $zw \in \mathbb{R} \wedge z \neq 0 \implies \exists a \in \mathbb{R}$ st $w = a\overline{z}$

assume $zw \in \mathbb{R}$ and $z \neq 0$, let $z = re^{i\theta}$ $\quad w = se^{i\varphi}$

$$\begin{aligned}
\therefore zw &= rse^{i(\theta+\varphi)} = rs & zw \in \mathbb{R} &\implies i = 0 \\
w &= \frac{rs}{z} = \frac{rs\overline{z}}{|z|^2} & z \neq 0 \quad \tfrac{1}{z} &= \tfrac{\overline{z}}{|z|^2} \\
&= \frac{rs\overline{z}}{r^2} = \frac{s}{r}\overline{z} = az \quad \blacksquare & z = ne0 &\implies r \neq 0
\end{aligned}$$

1.7 Euler $\times$ De Moivre's

Theorem. Euler's Formula

$$\begin{aligned}
e^{i\theta} &= \cos\theta + i\sin\theta \\
\implies e^{in\theta} &= \cos(n\theta) + i\sin(n\theta)
\end{aligned}$$

Theorem. De Moivre's

$$\begin{aligned}
(e^{i\theta})^n &= e^{in\theta} \\
\implies (\cos\theta + i\sin\theta)^n &= \cos(n\theta) + i\sin(n\theta) \qquad \text{from Euler's}
\end{aligned}$$

Example. $\cos 2\theta = \cos^2\theta - \sin^2\theta \quad \sin 3\theta = 3\cos^2\theta\sin\theta - \sin^3\theta$

$$\begin{aligned}
\cos 2\theta &= Re(\cos 2\theta + i\sin 2\theta) \\
&= Re((\cos\theta + i\sin\theta)^2) && \text{De Moivre} \\
&= Re(\cos^2\theta - \sin^2\theta + 2i\sin\theta\cos\theta) \\
&= \cos^2\theta - \sin^2\theta \\
\sin 3\theta &= Im(\cos 3\theta + i\sin 3\theta) \\
&= Im((\cos\theta + i\sin\theta)^3) && \text{De Moivre} \\
&= 3\cos^2\theta\sin\theta - \sin^3\theta
\end{aligned}$$

1.8 Roots

Theorem. Roots

- if $z, q \in \mathbb{C}, n \in \mathbb{Z}^+$ and $z^n = q$, then z is **nth root** of q
- $z^n = r^n e^{ni(\theta + 2k\pi)}$ has n solutions for $0 \leq k \leq n$

Example. Solve $z^6 = -64$

let $z = re^{i\theta}$

$$\begin{aligned}
z^6 &= (re^{i\theta})^6 = r^6 e^{6i\theta} = 64e^{i\pi} \\
\therefore r^6 &= 64 \implies r = 2 \\
6\theta &= \pi + 2\pi k \quad 0 \leq k \leq 6 \\
\theta &= \frac{\pi + 2\pi k}{6} \\
\therefore z &= 2e^{i(\pi/6)}, 2e^{i(3\pi/6)}, 2e^{i(5\pi/6)}, 2e^{i(7\pi/6)}, 2e^{i(9\pi/6)}, 2e^{i(11\pi/6)}
\end{aligned}$$

2 Systems of Linear Equations

Definition. System of m Equations with n Variables

$$A\vec{x} = \vec{b}$$

where

- $A \in \mathbb{R}^{m \times n}$ is the **coefficient** matrix
- $\vec{x} \in \mathbb{R}^n$ are the vairables
- $\vec{b} \in \mathbb{R}^m$ is the **constant vector**
- $\text{rank}\begin{bmatrix} A & b \end{bmatrix} = \text{rank}(A) \implies$ system is consistent
- A is invertible $\implies$ system has unique solution $\forall \vec{b}$

2.1 Gaussian Elimination

2.1.1 Elementary Row Operations

Definition.

1. interchange two rows `swap(ai, aj)`
2. multily a row by a nonzero number `ai *= n`
3. add a multiple of one row to another `ai += n * aj`

2.1.2 Row-Echelon Form

Definition. REF

- pivot $\equiv$ leading 1
- each row has pivot $\oplus$ is all 0
- each pivot is to the right of all pivots in the rows above it
- all zero-rows are at the bottom

Example. Matrices in REF

(I can't draw the stairs)

$$\begin{bmatrix} 1 & * & * \\ 0 & 1 & * \end{bmatrix} \begin{bmatrix} 0 & 1 & * & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & * & * \\ 0 & 1 & * \\ 0 & 0 & 1 \end{bmatrix}$$

Definition. Reduced Row-Echelon Form (RREF)

- is in REF
- all entries above a pivot is 0

2.1.3 Gaussian Elimination

Algorithm.

```
for c, col in a.cols.enumerate()
    for r, row in a.rows.enumerate()
        if col[r] == 0
            continue
        r /= a[r,c] // make pivot
        move row to top
    for r, row in a.rows.enumerate().skip(1)
        row -= a[r,c] * a.top // make zeros
```

- if has $\left[\begin{array}{ccc|c} 0 & \cdots & 0 & 1 \end{array}\right]$, then is inconsistent

Example. No Solution

$$\left[\begin{array}{ccc|c} 3 & 1 & -4 & -1 \\ 1 & 0 & 10 & 5 \\ 4 & 1 & 6 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 0 & -1 & 34 & 16 \\ 0 & -1 & 34 & 19 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 0 & -1 & 34 & 16 \\ 0 & 0 & 0 & -3 \end{array}\right]$$

$0 = -3 \implies$ has no solution, is inconsistent

Example. Has Parameters (General Solution)

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 2 & -4 & 1 & 0 & 5 \\ 1 & -2 & 2 & -3 & 4 \end{array}\right] \sim \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & -3 & -6 & 3 \\ 0 & 0 & 3 & -6 & 3 \end{array}\right] \sim \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \sim \left[\begin{array}{cccc|c} 1 & -2 & 0 & 1 & 2 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

no pivots for x_2 and x_4 $\implies x_2 = s \quad x_4 = t$

$$\begin{aligned} x_1 &= 2 + 2s - t \\ x_2 &= s \\ x_3 &= 1 + 2t \\ x_4 &= t \end{aligned}$$

for $s, t \in \mathbb{R}$

Example. Has Conditions

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & a \\ -1 & -2 & 1 & b \\ 3 & 7 & -1 & c \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 3 & 1 & a \\ 0 & 1 & 2 & a+b \\ 0 & -2 & -4 & c-3a \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & -5 & -2a-3b \\ 0 & 1 & 2 & a+b \\ 0 & 0 & 0 & c-a+2b \end{array}\right]$$

$0 = c - a + 2b \implies c = a - 2b$

if $c = a - 2b$, then

$$\begin{aligned} x_1 &= 5t - 2a - 3b \\ x_2 &= a + b - 2t \\ x_3 &= t \end{aligned}$$

for $t \in \mathbb{R}$

2.1.4 Rank of a Matrix

Definition.

number of pivots after being Gaussian Eliminated

Theorem.

if a $m \times n$ (m rows $\times n$ cols) matrix is **consistent**, and it has rank r, then

- the solution has $n - r$ parameters
- if $r < n$, has ∞ solutions
- if $r = n$ has a unique solution

2.2 Homogeneous Equations

Definition.

$$A\vec{x} = \vec{0}$$

- $x_i = 0$ is the **trivial solution**
- if parameter $\exists$, the system has a general **non trivial solution** ($\therefore$ inf solutions)

Theorem. $A\vec{x} = \vec{0}$ for $A \in \mathbb{R}^{m \times n}$

if $m < n$, then it has a nontrival solution

Proof. suppose a RREF augmented matrix A has m rows and n cols, $n > m$

suppose A has r leading vars $(0 \le r \le m)$

$\therefore A$ has $n - r$ non leading vars (parameters)

$\because m < n$

$\therefore 0 \le r < n$

$\therefore n - r > 0$

$\therefore A$ has parameters ■

2.2.1 Solution as Linear Combination

Definition.

suppose $\vec{x}, \vec{y}, \vec{z} \in \mathbb{R}^n$, $\vec{z}$ is a **linear combination** of x, y if

$$\vec{z} = s\vec{x} + t\vec{y}$$

for $s, t \in \mathbb{R}$

- $\vec{0}$ is a linear combination of all vectors

Example. Find general solution as linear comb of basic solutions

$$A = \begin{bmatrix} 1 & -2 & 3 & -2 \\ -3 & 6 & 1 & 0 \\ -2 & 4 & 4 & -2 \end{bmatrix}$$

1. convert to RREF augmented

$$\left[\begin{array}{cccc|c} 1 & -2 & 3 & -2 & 0 \\ -3 & 6 & 1 & 0 & 0 \\ -2 & 4 & 4 & -2 & 0 \end{array}\right] \sim \left[\begin{array}{cccc|c} 1 & -2 & 0 & -1/5 & 0 \\ 0 & 0 & 1 & -3/5 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

2. general solution

$$x_1 = 2s + 1/5t \quad x_2 = s \quad x_3 = 3/5t \quad x_4 = t$$

$$\vec{x} = \begin{bmatrix} 2s + 1/5t \\ s \\ 3/5t \\ t \end{bmatrix} = s \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1/5 \\ 0 \\ 3/5 \\ 1 \end{bmatrix} = s\vec{x}_1 + t\vec{x}_2$$

3. $\vec{x}_1$ and $\vec{x}_2$ are basic solutions

2.2.2 Solution of Any System as $\vec{x}_p + \vec{x}_0$

Algorithm. Solve $A\vec{x} = \vec{b}$ as $\vec{x}_p$ and $\vec{x}_0$

- convert to RREF
- $\vec{x}_p$ is solution when all parameters are 0
- $\vec{x}_0$ is the general solution when $\vec{b} = \vec{0}$

Example.

$$\left[\begin{array}{cccc|c} 2 & 1 & -1 & -1 & -1 \\ 3 & 1 & 1 & -2 & -2 \\ -1 & -1 & 2 & 1 & 2 \\ -2 & -1 & 0 & 2 & 3 \end{array}\right] \sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 3 \\ 0 & 1 & 0 & -4 & -9 \\ 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\vec{x}_p = \begin{bmatrix} 3 \\ -9 \\ -2 \\ 0 \end{bmatrix} \quad \vec{x}_0 = \begin{bmatrix} -1 \\ 4 \\ 1 \\ 0 \end{bmatrix} x_4$$

$$\vec{x} = \begin{bmatrix} 3 \\ -9 \\ -2 \\ 0 \end{bmatrix} + \begin{bmatrix} -1 \\ 4 \\ 1 \\ 0 \end{bmatrix} x_4$$

3 Matrix Algebra

Definition.

- $m \times n$ matrix means m rows and n cols
- a_{ij} means at row i and col j

3.1 Properties

3.1.1 Identity

Definition. Identity Matrix

$$I_1 = \begin{bmatrix} 1 \end{bmatrix} \quad I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad I_n = [e_{ij}] \text{ where } e_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

- $IA = AI = A$

Definition. Standard Basis of R^n

$$I_n = \begin{bmatrix} e_1 & e_2 & \cdots & e_n \end{bmatrix} \quad A = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}$$

- $Ae_i = a_i$
- useful for proofs in Mat-Vec Mul

3.1.2 (Skew) Symmetry

Definition.

- A is **symmetric** iif $A^T = A$
- A is **skew symmetric** iif $A^T = -A$

Example. Prove $\forall A \in \mathbb{R}^{n \times n} \exists$ symmetric S and skew symmetric W st $A = S + W$ and are uniquely determined by A

let $A = S + W$ where $S^T = S$ and $W^T = -W$

$$\therefore A^T = S^T + W^T = S - W$$

$$A + A^T = 2S \implies S = 1/2(A + A^T)$$

$$A - A^T = 2W \implies W = 1/2(A - A^T)$$

3.1.3 Idempotent

Definition.

for $A \in \mathbb{R}^{n\times n}$, A is **idempotent** iif

$$A^2 = A$$

Example. Prove $\forall A \in \mathbb{R}^{n\times n}$
P is idempotent $\implies Q = P + AP - PAP$ is idempotent

assume $P = P^2$

$$Q = P + AP - PAP$$

$$\because PQ = P^2 + PAP - P^2AP = P + PAP - PAP = P$$

$$\therefore Q^2 = PQ + APQ - PAPQ = P + AP - PAP = Q \quad \blacksquare$$

3.2 Add

Definition. `[m,n] * [m,n] → [m,n]`

if $A = [a_{ij}]$ and $B = [b_{ij}]$, then

$$A + B = [a_{ij} + b_{ij}]$$

- commutativity, associativity
- add-identity: 0_{mn}
- add-inverse: $-A$

3.3 Transpose

Definition. `[m,n]ᵀ → [n,m]`

$$A = [a_{ij}] \implies A^T = [a_{ji}]$$

Theorem.

- $(A^T)^T = A$
- $(kA)^T = kA^T$
- $(A + B)^T = A^T + B^T$
- $(AB)^T = B^T A^T$
- $\forall A \in \mathbb{R}^{1\times 1} \quad A^T = A$

3.4 Scalar Mul

Definition. `[m,n] * k → [m,n]`

if $A = [a_{ij}]$ and $k \in \mathbb{R}$, then

$$kA = [ka_{ij}]$$

- commutativity, associativity, distributivity
- mul-identity: 1

3.5 Matrix-Vector Mul

Definition. Matrix written as columns

a $m \times n$ matrix can be written as

$$A = \begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n \end{bmatrix}$$

where $\vec{a}_i \in \mathbb{R}^m$

Definition. `[m,n] * [n] → [m]`

if $m \times n$ matrix $A = \begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n \end{bmatrix}$ and $\vec{x}$ is any n-vec, then

$$A\vec{x} = x_1\vec{a}_1 + x_2\vec{a}_2 + \ldots + x_n\vec{a}_n$$

- associativity, distributivity

Example. Prove $\forall \vec{x} \in \mathbb{R}^n \quad A\vec{x} = B\vec{x} \implies A = B$

for $A, B \in \mathbb{R}^{m\times n}$

$$\text{let } A = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}$$
$$B = \begin{bmatrix} b_1 & b_2 & \cdots & b_n \end{bmatrix}$$
$$\{e_i\} \text{ be standard basis of } \mathbb{R}^n$$

assume $\forall \vec{x} \in \mathbb{R}^n \quad A\vec{x} = B\vec{x}$

$$\therefore \forall 1 \le i \le n \quad Ae_i = Be_i \implies a_i = b_i \implies A = B \quad \blacksquare$$

3.5.1 System of Equations

Theorem. let $A\vec{x} = \vec{b}$ be a system of equations and rank $A = r$

- rank $\begin{bmatrix} A \mid b \end{bmatrix} = r \oplus r + 1$
- system is **consistent** iff rank is r
- system is **inconsistent** iff rank is $r + 1$

3.6 Matrix-Matrix Mul

Definition. `[m,n] * [n,k] → [m,k]`

if $A \in \mathbb{R}^{m\times n}$ and $B = \begin{bmatrix} b_1 & b_2 & \ldots & b_n \end{bmatrix} \in \mathbb{R}^{n\times k}$, then $AB \in \mathbb{R}^{m\times k}$ and

$$AB = A\begin{bmatrix} b_1 & b_2 & \ldots & b_n \end{bmatrix} = \begin{bmatrix} Ab_1 & Ab_2 & \ldots & Ab_n \end{bmatrix} = [c_{ij}]$$

where $c_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}$

Theorem. Assume $a \in \mathbb{R} \quad A, B, C \in \mathbb{R}^{m\times n}$

- $(AB)^T = B^T A^T$
- associativity, distributivity

$$A(BC) = (AB)C$$
$$a(AB) = (aA)B = A(aB)$$
$$ABCD \not\equiv ACDB$$

- NO commutativity: $AB \not\equiv BA$
  A **commutes** with B iff $AB = BA$

Example. Prove $(AB)^T = B^T A^T$

let $A = [a_{ij}]$ and $B = [b_{ij}]$

$$\therefore A^T = [a'_{ij}] \quad B^T = [b'_{ij}] \text{ where } a'_{ij} = a_{ji} \quad b'_{ij} = b_{ji}$$

let $B^T A^T = [c_{ij}]$

$$\therefore c_{ij} = \sum_{k=1}^{n} b'_{ik}a'_{kj} = \sum_{k=1}^{n} b_{ki}a_{jk} = \sum_{k=1}^{n} a_{jk}b_{ki} = (AB)_{ji} = (AB)^T_{ij}$$

Example. let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times n}$
Prove $A \neq 0 \wedge AB = 0 \implies \nexists C \in \mathbb{R}^{n\times n}$ st $BC = I$

assume $A \neq 0$ and $AB = 0$, let $C \in \mathbb{R}^{n\times n}$

$$BC = I \implies 0 = 0C = (AB)C = A(BC) = AI = A$$

which contradicts the initial assumption ■

3.6.1 Calculate Product

Definition. Dot Product

let $a, b \in \mathbb{R}^n$

$$a \cdot b = \sum a_i b_i$$

Algorithm. AB

represent A as rows a_i and B as cols b_i

$$\begin{array}{c|c} & B \\ \hline A & AB \end{array} = \begin{array}{c|cccc} & b_1 & b_2 & \cdots & b_n \\ \hline a_1 & a_1 \cdot b_1 & a_1 \cdot b_2 & \cdots & a_1 \cdot b_n \\ a_2 & a_2 \cdot b_1 & a_2 \cdot b_2 & \cdots & a_2 \cdot b_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_m & a_m \cdot b_1 & a_m \cdot b_2 & \cdots & a_m \cdot b_n \end{array}$$

Example. Prove $\forall B \in \mathbb{R}^{2\times 2}$ $A = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$ commutes with B

let $B = \begin{bmatrix} b & c \\ d & e \end{bmatrix}$ for $b, c, d, e \in \mathbb{R}$

$$\begin{array}{cc|cc} & & b & c \\ & & d & e \\ \hline a & 0 & ab & ac \\ 0 & a & ad & ae \end{array} \qquad \begin{array}{cc|cc} & & a & 0 \\ & & 0 & a \\ \hline b & c & ab & ac \\ d & e & ad & ae \end{array}$$

$$\therefore AB = BA = \begin{bmatrix} ab & ac \\ ad & ae \end{bmatrix} \quad ■$$

3.7 Trace

Definition.

$$\operatorname{tr}(A) = \operatorname{tr}[a_{ij}] = \sum a_{ii}$$

Theorem.

- $\operatorname{tr}(A + B) = \operatorname{tr}(A) + \operatorname{tr}(B)$
- $\operatorname{tr}(kA) = k\operatorname{tr}(A)$
- $\operatorname{tr}(A^T) = \operatorname{tr}(A)$
- $\operatorname{tr}(AB) = \operatorname{tr}(BA)$
- $\operatorname{tr}(AB) \not\equiv \operatorname{tr}(A)\operatorname{tr}(B)$
- $\operatorname{tr}(A^T B) = \operatorname{tr}(AB^T) = \sum a_{ii} b_{ii}$

Example. Prove $\forall A \in \mathbb{R}^{m\times n}, B \in \mathbb{R}^{n\times m}$ $\quad \operatorname{tr}(AB) = \operatorname{tr}(BA)$

let $AB = C$

$$\because C_{ii} = \sum_{j=1}^{n} A_{ij} B_{ji}$$

$$\therefore \operatorname{tr}(AB) = \sum_{i=1}^{n} C_{ii} = \sum_{i=1}^{m}\sum_{j=1}^{n} A_{ij} B_{ji}$$

$$= \sum_{j=1}^{n}\sum_{i=1}^{n} A_{ji} B_{ij} = \operatorname{tr}(BA)$$

3.8 Inverse

Definition.

if $A \in \mathbb{R}^{n\times n}$, then A **is invertible** iff any:

- A^T is invertible
- $\exists B$ st $AB = I$ or $\exists C$ st $CA = I$
- $\det A \neq 0$
- $A = \prod E_i \equiv A \sim I$
  i.e. $A \to I$ is possible by row operations
- $\ker(A) = \{0\}$
  i.e. $A\vec{x} = 0$ has only the solution $\vec{x} = 0$
- $\forall \vec{b} \quad A\vec{x} = \vec{b}$ has solutions
- $\operatorname{rank}(A) = n$

if $AB = I = CA$ for all square matrices, then

$$B = C = A^{-1}$$

Theorem. Assume A is invertible/has an inverse

- $(A^T)^{-1} = (A^{-1})^T$
- $(A_1 A_2 \cdots A_n)^{-1} = A_n^{-1} \cdots A_2^{-1} A_1^{-1}$
- $(A^k)^{-1} = (A^{-1})^k$
- $(aA)^{-1} = {}^1\!/_a A^{-1}$ for $a \neq 0$

Algorithm. Find Inverse

use Gaussian Algorithm to

$$[\ A \mid I\ ] \to [\ I \mid A^{-1}\ ]$$

Example. Prove if A, B are invertible, then $(AB)^{-1} = B^{-1}A^{-1}$

if $(L)^{-1} = R$, show $LR = I = RL$

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I \quad ■$$

Example. Prove AB is invertible $\implies$ A and B are both invertible

assume $AB \in \mathbb{R}^{n\times n}$ is invertible

$$\therefore \exists C \in \mathbb{R}^{n\times n} \text{ st } (AB)C = C(AB) = I$$

$$A(BC) = I \implies A \text{ is invertible}$$

$$(CA)B = I \implies B \text{ is invertible}$$

3.8.1 Solution of a System

Theorem. Find Unique Solution of a System

if a system of n eqns and n vars is written as

$$A\vec{x} = \vec{b}$$

then

$$A \text{ is invertible } \implies \vec{x} = A^{-1}\vec{b}$$

3.9 Elementary Matrices

Definition.

an $n \times n$ matrix E is **elementary** if E can be produced from a single elementary row operation on I

Theorem. Lemma

suppose an elementary operation op() and $A \in \mathbb{R}^{m \times n}$

- $\text{op}(A) = \text{op}(I_m)A = EA$
- $\text{op}(E^{-1}) = I$ where E^{-1} is also elementary and is the reverse operation

3.9.1 Theorem

Theorem. Suppose $A \in \mathbb{R}^{m \times n}$ and $A \to R$ by elem row ops

- $A \to R \equiv \begin{bmatrix} A & I_m \end{bmatrix} \to \begin{bmatrix} R & U \end{bmatrix}$
- $R = UA$ for $U \in \mathbb{R}^{m \times m}$ is invertible
- $U = E_n \cdots E_2 E_1$ where E_i is the ith row operation
- $B \in \mathbb{R}^{n \times n}$ is invertible iff $B = \prod E_i$
- $A = UR \implies A \overset{r}{\sim} R$ **row equivalent**

Example. Prove $A \overset{r}{\sim} B \Rightarrow B \overset{r}{\sim} A$ and $A \overset{r}{\sim} B \overset{r}{\sim} C \Rightarrow A \overset{r}{\sim} C$

1. assume $A \overset{r}{\sim} B$

$$\therefore A = UB \text{ for invertible } U$$
$$\therefore B = U^{-1}A \implies B \overset{r}{\sim} A \quad \blacksquare$$

2. assume $A \overset{r}{\sim} B \overset{r}{\sim} C$

$$\therefore A = UB \quad B = VC \text{ for invertible } U, V$$
$$\therefore A = UVC = (UV)C \implies A \overset{r}{\sim} C \quad \blacksquare$$

Example. let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times m}$
Prove $m > n \implies AB$ is not invertible

$$m > n \implies B\vec{x} = 0 \text{ has nontrivial solution}$$
$$\therefore \exists \vec{x} \neq 0 \text{ st } B\vec{x} = 0$$
$$\therefore A(B\vec{x}) = AB\vec{x} = 0 \text{ has nontrivial solution}$$
$$\therefore AB \text{ is not invertible} \quad \blacksquare$$

3.10 Linear Map

Definition. $\forall x, y \in \mathbb{R}^n$

- if T is linear, then $T(x) = T_A(x) = Ax$ where

$$A = \begin{bmatrix} T(e_1) & T(e_2) & \cdots & T(e_n) \end{bmatrix}$$

- $T(x + y) = T(x) + T(y)$
- $T(ax) = aT(x)$
- $T = A \wedge S = B \implies (T \circ S)(x) = (AB)x$

Theorem. Linearity

if $T : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation, then

$$T\left(\sum a_i \vec{x_i}\right) = \sum a_i T(\vec{x_i})$$

proven using induction

Definition. Kernel and Image for $T : \mathbb{R}^n \to \mathbb{R}^m$

$$\ker(T) = \{x \in \mathbb{R}^n : T(x) = 0\}$$
$$\text{im}(T) = \{T(x) : x \in \mathbb{R}^n\}$$

- $x_1, x_2 \in \ker(T) \implies ax_1 + bx_2 \in \ker(T)$
- $x_1, x_2 \in \text{im}(T) \implies ax_1 + bx_2 \in \text{im}(T)$

Theorem. Geometrical Linear Transformatins

they are all $T : \mathbb{R}^2 \to \mathbb{R}^2$

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \qquad \text{(rotate anticlock-wise)}$$

$$Q_m = \frac{1}{1+m^2}\begin{bmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{bmatrix} \qquad \text{(reflect on } y = mx\text{)}$$

3.11 Practice

Example. let $A \in \mathbb{R}^{n\times n}$ st $\vec{x} \neq 0 \implies \vec{x}^T A\vec{x} > 0$
Show $\forall$skew-symmetric $B \in \mathbb{R}^{n\times n}, \vec{b} \in \mathbb{R}^n$

$$(A+B)\vec{x} = \vec{b}$$

as a unique solution

$\because (A+B)\vec{x} = \vec{b}$ has a unique solution iff $(A+B)\vec{x} = 0 \implies \vec{x} = 0$
$(\vec{x} \neq 0 \implies \vec{x}^T A\vec{x} > 0) \equiv (\vec{x}^T A\vec{x} \leq 0 \implies \vec{x} = 0)$
$\therefore$ given $\vec{x}^T A\vec{x} \leq 0 \implies \vec{x} = 0$ and $B^T = -B$
show $(A+B)\vec{x} = 0 \implies \vec{x} = 0$

1. show $\vec{x}^T B\vec{x} = 0$

$$\begin{aligned} \vec{x}^T B\vec{x} &= (\vec{x}^T B\vec{x})^T && (A^T = A \quad \forall A \in \mathbb{R}^{1\times 1}) \\ \vec{x}^T B^T \vec{x} &= -\vec{x}^T B\vec{x} \\ \iff \vec{x}^T B\vec{x} &= -\vec{x}^T B\vec{x} \\ 2\vec{x}^T B\vec{x} &= 0 \\ \vec{x}^T B\vec{x} &= 0 \end{aligned}$$

2. show $(A+B)\vec{x} = 0 \implies \vec{x} = 0$

$$\begin{aligned} (A+B)\vec{x} &= A\vec{x} + B\vec{x} = 0 && \text{(distributivity)} \\ \vec{x}^T A\vec{x} + \vec{x}^T B\vec{x} &= 0 && \text{(left mul by } \vec{x}^T) \\ \vec{x}^T A\vec{x} &= 0 && \text{(from 1)} \\ \vec{x}^T A\vec{x} \leq 0 &\implies \vec{x} = 0 && \text{(given)} \end{aligned}$$

4 Eigenvectors

4.1 Cofactor Expansion

Definition. $\det(A)$ for $A \in \mathbb{R}^{2\times 2}$ and $B \in \mathbb{R}^{3\times 3}$

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$\det(B) = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a\begin{vmatrix} e & f \\ h & i \end{vmatrix} - b\begin{vmatrix} d & f \\ g & i \end{vmatrix} + c\begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

Definition. Cofactor expansion of $\det(A)$ along row 1

for $A = [a_{ij}] \in \mathbb{R}^{n\times n}$

$$\det(A) = |A| = \sum a_{1i}c_{1i}(A)$$

$$c_{ij}(A) = (-1)^{i+j}|A_{ij}|$$

A_{ij} is the matrix of A after removing row i and col j

Theorem. let $A \in \mathbb{R}^{n\times n}$ and $u \in \mathbb{R}$

- if $\exists$ a row/col is a **linear combination** of any other rows/cols then $|A| = 0$
- if B is A but 2 rows/cols are swapped then $|B| = -|A|$
- if B is A but a row/col `*= c` then $|B| = c|A|$
- `A_i += c` $\implies c|A|$
- `A_i += u * A_j` for `i` $\neq$ `j` $\implies$ **keeps** $|A|$

4.2 Determinant and Matrix Inverse

4.2.1 Determinant

Theorem.

- $|I| = 1$
- $|cA| = c^n|A|$
- $|A^T| = |A|$ for $A \in \mathbb{R}^{n\times n}$
- $|AB| = |A||B|$
- $|A^{-1}| = \frac{1}{|A|}$ for invertible A

Example. find $|A|$ if $A^3 = A$

$$A^3 = A \implies |A^3| = |A| \implies |A|^3 - |A| = 0$$

$$|A|(|A|^2 - 1) = 0$$

$$\therefore |A| = 0, \pm 1$$

4.2.2 Adjugate

Definition.

$$\operatorname{adj}(A) = [c_{ij}(A)]^T$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \implies \operatorname{adj}(A) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Theorem.

- $A(\operatorname{adj} A) = |A|I = (\operatorname{adj} A)A$
- $A^{-1} = \frac{1}{|A|}\operatorname{adj} A$

4.3 Eigenvalue

Definition. λ

λ is **an eigenvalue** of $A \in \mathbb{R}^{n\times n}$ if for $\vec{x} \neq 0 \in \mathbb{R}^n$

$$A\vec{x} = \lambda\vec{x} \equiv |\lambda I - A| = 0$$

and $\vec{x}$ is an λ-eigenvector

Definition. $c_A(\lambda)$

the **characteristic polynomial** of A is

$$c_A(\lambda) = |\lambda I - A| = (\lambda - \lambda_1)(\lambda - \lambda_2)\cdots(\lambda - \lambda_n)$$

- **eigenvalues** of A is λ when

$$c_A(\lambda) = 0$$

- λ**-eignvectors** is $\vec{x} \neq 0$ when

$$(\lambda I - A)\vec{x} = 0$$

4.3.1 A-Invariance

Definition. for $A \in \mathbb{R}^{n\times n}$ and $\vec{x} \neq 0 \in \mathbb{R}^n$

- $L_x = \{\lambda\vec{x} : \lambda \in \mathbb{R}\}$ is A**-invariant** if $\vec{x} \in L \implies A\vec{x} \in L$
- $\vec{x}$ is an eigenvec of A iff L_x is A-invariant

4.3.2 Multiplicity

Definition.

an eigenval λ has **multiplicity** n if it occurs n items as root of $c_A(x)$

- $(\lambda - 1)^2 \implies \lambda_1 = 1$ with multiplicity 2

4.3.3 Diagonalisation

Definition. Power of Matrix

if $A = PDP^{-1}$ then $A^k = PD^kP^{-1}$ for $k \in \mathbb{Z}^+$

Definition. Diagonal Matrix

$$D = \operatorname{diag}(\lambda_1, \lambda_2, \cdots, \lambda_n) = \begin{bmatrix} \lambda_1 e_1 & \lambda_2 e_2 & \cdots & \lambda_n e_n \end{bmatrix}$$

Definition.

$A \in \mathbb{R}^{n\times n}$ is **diagonalisable** if

$$\exists P \in \mathbb{R}^{n\times n} \text{ st } P^{-1}AP = D$$

where the invertible P is a **diagonalising matrix** of A

Theorem.

for $A \in \mathbb{R}^{n\times n}$

- A is diagable iff A has eigenvecs $\vec{x}_i$ st $P = \begin{bmatrix} \vec{x}_i \end{bmatrix}$ is invertible
- if diagable, $P^{-1}AP = \operatorname{diag}(\lambda_1, \cdots, \lambda_n)$ where $A\vec{x}_i = \lambda_i\vec{x}_i$

Algorithm. Diagonise $A \in \mathbb{R}^{n\times n}$

1. find distinct eigenvals λ of A
2. for each λ, find basic eigenvec $\vec{x}$ from $(\lambda I - A)\vec{x} = 0$
3. A is diagable iff $\exists n$ eigenvecs
4. if diagable, then $P = \begin{bmatrix} \vec{x}_i \end{bmatrix}$ st $P^{-1}AP$ is diagonal